Protocol: Protocol I8B-MC-ITSL(a) A Mixed Meal Tolerance Test Study to Evaluate the Pharmacokinetics and Pharmacodynamics of LY900014 Compared to Humalog Following a Single Dose in Adults with Type 1 Diabetes

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LY900014

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1. Protocol Synopsis

Title of Study:
A Mixed Meal Tolerance Test Study to Evaluate the Pharmacokinetics and Pharmacodynamics of LY900014 Compared to Humalog Following a Single Dose in Adults with Type 1 Diabetes

Rationale:
This study in patients with type 1 diabetes mellitus (T1DM) aims to compare the pharmacokinetic (PK) profiles of insulin lispro after administration of LY900014 and Humalog and their impact on postprandial blood glucose during a standardised mixed meal tolerance test (MMTT). Furthermore, this study plans to characterise the PK and pharmacodynamics (PD) of insulin aspart analogues NovoRapid® and FIASP® in patients with T1DM under identical study conditions. A non-diabetic healthy subject cohort is included in the study to characterise "normal" insulin secretory and glucose response during an identical standardised meal. Healthy subjects will not receive any of the study drugs.

Evaluation of these meal-time insulins in the context of a standardised test meal cannot be used to infer clinical efficacy with the same rigour as a glycated haemoglobin assessment following long-term treatment. However, assessment of these analogue insulin formulations under identical conditions provides a unique opportunity to obtain comparative information on the PK and PD in a setting appropriate for evaluation of meal-time insulin.

Objective(s)/Endpoints:

Objectives	Endpoints
Primary 1. To evaluate the differences in PK between LY900014 and Humalog following a single dose in patients with T1DM	1. Early 50% t_{max} and AUC(0-30min)
Secondary 1. To characterise the PK profiles of LY900014, Humalog, NovoRapid and FIASP following a single dose in patients with T1DM 2. To evaluate the differences in PD between LY900014 and Humalog following a single dose in patients with T1DM 3. To characterise the PD response to LY900014, Humalog, NovoRapid and FIASP, as assessed by the MMTT, in patients with T1DM	1. Early 50% t_{max} 2. ΔAUC(0-1h), ΔAUC(0-5h) 3. ΔAUC(0-1h), ΔAUC(0-5h)

Abbreviations: AUC(0-30min) = area under the concentration versus time curve from time 0 to 30 minutes; ΔAUC = area under the baseline subtracted glucose concentration versus time curve; ΔAUC(0-1h) = ΔAUC from time 0 to 1 hour postmeal; ΔAUC(0-5h) = ΔAUC from time 0 to 5 hours postmeal; early 50% t_{max} = time to early half-maximal drug concentration; MMTT = mixed meal tolerance test; PD = pharmacodynamics; PK = pharmacokinetics; T1DM = type 1 diabetes mellitus.

Summary of Study Design:

Study I8B-MC-ITSL is a clinical pharmacology, patient- and investigator-blind, randomised, 4-treatment, 4-period crossover study in patients with T1DM. The study also involves healthy subjects who will undergo a single MMTT but will not receive any study drug.

Treatment Arms and Planned Duration for an Individual Patient with T1DM:
Patients will be screened over a 14-day period prior to start of a 7- to 14-day lead-in (insulin transition) period. Patients will then participate in a dose-finding assessment (1 to 7 days prior to MMTT in Period 1) and subsequently randomised to 1 of 4 treatment sequences according to the actual randomisation table provided to the clinical research unit (CRU). In each study period, patients will undergo a MMTT. Each patient will receive a single individualised subcutaneous (SC) injection of LY900014, Humalog, FIASP or NovoRapid immediately before the standardised test meal. All treatment procedures will be inpatient stays of approximately 2 days per period and require a maximum of 6 weeks to complete 4 periods. Each study dosing will be separated by a minimum of 21 hours and may occur on consecutive days; however, patient inclusion is subject to required screening criteria. The follow-up or early discontinuation visit should occur at least 14 days after the last dose of the study drug.

Treatment Arms and Planned Duration for an Individual Healthy Subject:

Subjects will be screened over a 28-day period prior to the MMTT. Subjects will be admitted to the CRU the evening before the MMTT. Each subject will receive a standardised test meal on Day 1. Subjects will remain inpatient in the CRU for 2 days.

Treatment (applicable to patients):

For a given patient, the single individualised dose will be the same for all the treatments.

LY900014: Single individualised SC dose per assessment period

Humalog: Single individualised SC dose per assessment period

FIASP: Single individualised SC dose per assessment period

NovoRapid: Single individualised SC dose per assessment period

Number of Patients and Subjects:
Up to 74 patients with T1DM may be enrolled so that approximately 64 patients complete the study. Replacement patients will adopt all 4 assigned crossover treatments of the original patient's randomisation schedule.

For the healthy subject cohort, sufficient subjects will be enrolled to have 12 healthy subjects complete the study.

Statistical Analysis:

Primary statistical analyses of PK and PD parameters will be conducted on the set of patients who complete all treatment periods with identical prandial insulin does for the MMTTs during the study. Supportive analyses may be done on the key parameters for the patients who complete at least 2 treatment periods. Safety analyses will be conducted on the set of patients receiving at least 1 dose of the study drug to which they are randomised, regardless of whether or not they completed all protocol requirements.

Unless otherwise noted, all tests of treatment effects will be conducted at a 2-sided alpha level of 0.05, and confidence intervals (CIs) will be calculated at 95%, 2-sided. No multiplicity test adjustment will be made for secondary objectives.

Safety: All investigational product- and protocol procedure-related adverse events will be listed, and if the frequency of events allows, safety data will be summarised using descriptive methodology. Safety parameters will be listed and summarised using standard descriptive statistics.

Pharmacokinetic:

The mixed-effects model will be used for the analysis of the PK time parameters (time to early half-maximal drug concentration [early 50% t_{max}], time to late half-maximal drug concentration [late 50% t_{max}] and time to maximum observed drug concentration [t_{max}]) without log transformation. Least-squares means (LSmeans), treatment differences in LSmeans and the corresponding 95% CIs for the treatment differences will be estimated from the model. The p-value on the difference between LSmeans will be used to determine statistical significance. The treatment ratios and 95% CIs for the ratios will be calculated using Fieller's theorem.

Log-transformed area under the concentration versus time curves (AUCs), maximum observed drug concentration (C_{max}) for insulin lispro/aspart may be evaluated to estimate geometric means, ratios of geometric means and their corresponding 95% CIs using the same mixed-effect model as PK time parameters.

Pharmacodynamics:

Data will be analysed for the patients during each MMTT. The change from baseline values (the average of -30, -15 and 0 minutes) represented as the 0-hour time point following the start of the MMTT for each patient will be calculated. The area under the baseline subtracted glucose concentration versus time curve from time 0 to 1 hour post-meal (ΔAUC[0-1h]), area under the baseline subtracted glucose concentration versus time curve from time 0 to 2 hours post-meal (ΔAUC[0-2h]) and area under the baseline subtracted glucose concentration versus time curve from time 0 to 5 hours post-meal (ΔAUC[0-5h]) will be calculated. In addition, the change from baseline maximum glucose observed during the 5 hours post-meal and change from baseline 1 hour glucose and 2 hour glucose after the start of the meal will be calculated. Other partial ΔAUCs may be calculated, as deemed appropriate.

Summary statistics will be presented by treatment. All PD parameters (including ΔAUCs, glucose changes from baseline and time parameters) on the original scale will be analysed using the mixed-effects model that includes treatment, sequence and period as fixed effects and patient within sequence as a random effect. The p-value on the difference between LS means will be used to determine statistical significance and the corresponding 95% CIs for the LSmean ratios from Fieller's theorem will be presented.

2. Schedule of Activities

Study Schedule Protocol I8B-MC-ITSL for Patients with T1DM

	Screening	Lead-In	Dose-Finding Assessment	Study Period (Periods 1 to 4)		Follow-up/Early Discontinuation	Comments
Procedure	**Up to 14 days prior to lead-in**	**7-14 days prior to dose-finding assessment**	**1-7 days prior to Period 1, Day 1**	**Day -1**	**Day 1**	**≥14 days after discharge**	
Informed consent	X						At least 1 day before screening procedures. Screening procedures should take place no later than 28 days after signing the informed consent.
Medical history and physical examination	X						
Height	X						At screening only.
Weight	X			X		X	Period 1 only.
Vital signs (supine): blood pressure and pulse rate	X				Predose and 420 minutes postdose	X	
12-lead ECG (supine)	X				Predose (Period 1)	X	Single ECGs will be collected for safety.
Clinical laboratory tests	X		X		Predose for Period 1 only	X	Fasting laboratory test. All clinical laboratory tests will be analysed at a local laboratory (see Section 9.4.2).
Alcohol breath test				X			Alcohol breath test to be done at admission for all periods.

	Screening	Lead-In	Dose-Finding Assessment	Study Period (Periods 1 to 4)		Follow-up/Early Discontinuation	Comments
Procedure	**Up to 14 days prior to lead-in**	**7-14 days prior to dose-finding assessment**	**1-7 days prior to Period 1, Day 1**	**Day -1**	**Day 1**	**≥14 days after discharge**	
Lead-in/insulin transition activities		X					Once patients have completed screening procedures, they will switch from prescribed basal insulin to site-provided insulin glargine. Patients will receive general diabetes training. A patient diary will be provided for recording dosing and other required information (Section 5.1.1.1). Lead-in period may be extended by 2 days as needed.
Patient admission to CRU				X			Admitted to CRU on the evening of Day -1 for all study periods. If the dose-finding assessment occurs from Day -2, the patient may continue to remain inpatient in the CRU until the completion of study MMTT for Period 1.

	Screening	Lead-In	Dose-Finding Assessment	Study Period (Periods 1 to 4)		Follow-up/Early Discontinuation	Comments
Procedure	**Up to 14 days prior to lead-in**	**7-14 days prior to dose-finding assessment**	**1-7 days prior to Period 1, Day 1**	**Day -1**	**Day 1**	**≥14 days after discharge**	
Dose-finding MMTT			X				Patients will undergo dose-finding assessment with Humalog and a standardised test meal. Dose-finding MMTT will occur after the insulin transition. Dose-finding assessment should occur between 1 and 7 days prior to MMTT in Period 1. To take place between 0700 hours and 1100 hours. See Section 5.1.1.2 for rescheduling.
Randomisation				X			To take place before run-in of Period 1 Day 1.
Pregnancy test	X			X		X	For female patients only. Serum pregnancy test will be performed at screening and urine pregnancy tests will be performed at the time of admission in Periods 1 to 4 and at follow-up (see Appendix 2).
Standardised dinner				X			Approximately at 1900 hours. The evening dinner will be identical for every period.

	Screening	Lead-In	Dose-Finding Assessment	Study Period (Periods 1 to 4)		Follow-up/Early Discontinuation	Comments
Procedure	**Up to 14 days prior to lead-in**	**7-14 days prior to dose-finding assessment**	**1-7 days prior to Period 1, Day 1**	**Day -1**	**Day 1**	**≥14 days after discharge**	
Medical assessment				X	Predose and before discharge from CRU	X	Medical assessment includes medical review and targeted examination, and as appropriate review of concomitant medication, patient diary and MMTT exclusion criteria (see Section 6.2.1).
Glucose stabilisation/run-in					X		From approximately 7 hours to 30 minutes before dosing: IV infusion of glucose (dextrose solution) and/or insulin glulisine to achieve a target blood glucose level of 135±15 mg/dL (7.5±0.8 mmol/L). Blood glucose concentrations will be monitored at a minimum of 30-minute intervals (see Sections 9.6.1 and 5.1.1.3). See Section 6.2.3 for rescheduling.

	Screening	Lead-In	Dose-Finding Assessment	Study Period (Periods 1 to 4)		Follow-up/Early Discontinuation	Comments
Procedure	**Up to 14 days prior to lead-in**	**7-14 days prior to dose-finding assessment**	**1-7 days prior to Period 1, Day 1**	**Day -1**	**Day 1**	**≥14 days after discharge**	
Blood glucose sampling (MMTT)			20, 40, 60, 80, 100, 120, 140, 160, 180, 200, 220, 240, 260, 280, 300 minutes post-meal		-30, -15, 0 minutes pre-meal, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 110, 120, 135, 150, 165, 180, 195, 210, 225, 240, 300 minutes post-meal		0-minute time point sample to be taken at the start of the meal.
Study MMTT					X		Standardised test meal will be administered at approximately 0700 hours (with allowance up to 1100 hours) and should be consumed within 15 minutes (see Section 5.1.1.3). MMTT may occur on consecutive days.
Study drug administration					X		Immediately before the start of meal according to the randomisation schedule: Study drug will be administered at approximately the same time on Day 1 of Periods 1 to 4. There will be at least 21 hours between study doses.

	Screening	Lead-In	Dose-Finding Assessment	Study Period (Periods 1 to 4)		Follow-up/Early Discontinuation	Comments
Procedure	**Up to 14 days prior to lead-in**	**7-14 days prior to dose-finding assessment**	**1-7 days prior to Period 1, Day 1**	**Day -1**	**Day 1**	**≥14 days after discharge**	
Insulin lispro/insulin aspart PK sampling					0 (predose), 1, 2, 3, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 70, 90, 120, 150, 180, 240, 300, 360 and 420 minutes postdose		Sampling times are relative to the time of study drug administration in each period.
Discharge from CRU					X		Patients may be discharged after all study procedures are completed. Patients may be offered a meal following completion of all PK and PD assessments.

Abbreviations: CRU = clinical research unit; ECG = electrocardiogram; MMTT = mixed meal tolerance test; PD = pharmacodynamics; PK = pharmacokinetics.

Note: The CRU should schedule activities as appropriate. In cases where several study procedures are scheduled at the same time, the order of priority will be as follows: PK samples including sampling for blood glucose and laboratory samples per protocol nominal times. ECGs and vital sign measurements should be scheduled before but as close as possible to the PK sampling times.

Study Schedule Protocol I8B-MC-ITSL for Healthy Subjects

	Screening	Study Period		Comments
Procedure	Days -28 to -2	Day -1	Day 1	
Informed consent	X			At least 1 day before screening procedures. Screening procedures should take place no later than 28 days after signing the informed consent.
Medical history and physical examination	X			
Height	X			At screening only.
Weight	X	X		
Vital signs (supine): blood pressure and pulse rate	X			
12-lead ECG (supine)	X			Single ECGs will be collected for safety.
Clinical laboratory tests	X			Fasting laboratory test. Laboratory tests will be analysed at a local laboratory.
Oral glucose tolerance test	X			Plasma glucose measurements before ingestion of glucose beverage and at 1 and 2 hours after the start of ingestion of glucose beverage.
Subject admission to CRU		X		Admitted to CRU on the evening of Day -1.
Pregnancy test	X	X		For female subjects only. Serum pregnancy test will be performed at screening and urine pregnancy tests will be performed at admission to the CRU (see Appendix 2).
Standardised dinner		X		Approximately at 1900 hours.
Medical assessment		X	Before discharge from CRU	Medical assessment includes medical review and targeted examination, and as appropriate review of concomitant medication and MMTT exclusion criteria (see Section 6.2.2).
Blood glucose sampling (MMTT)			-30, -15, 0 minutes pre-meal, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 110, 120, 135, 150, 165, 180, 195, 210, 225, 240, 300 minutes post-meal	0-minute time point sample to be taken at the start of the meal.

	Screening	Study Period		Comments
Procedure	Days -28 to -2	Day -1	Day 1	
MMTT			X	Standardised test meal will be administered at approximately 0700 hours (with allowance up to 1100 hours) and should be consumed within 15 minutes (see Section 5.1.2).
Endogenous insulin sampling			0 (pre-meal), 1, 2, 3, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 70, 90, 120, 150, 180, 240, 300, 360 and 420 minutes postmeal	Sampling times are relative to the time of meal administered.
Discharge from CRU			X	Subjects may be discharged after all study procedures are completed. Subjects may be offered a meal following completion of all PK and PD assessments.

Abbreviations: CRU = clinical research unit; ECG = electrocardiogram; MMTT = mixed meal tolerance test; PD = pharmacodynamics; PK = pharmacokinetics.

Note: The CRU should schedule activities as appropriate. In cases where several study procedures are scheduled at the same time, the order of priority will be as follows: PK samples including sampling for blood glucose and laboratory samples per protocol nominal times. ECGs and vital sign measurements should be scheduled before but as close as possible to the PK sampling times.

3. Introduction

3.1. Study Rationale

A prandial insulin with faster-on and/or faster-off characteristics might reduce glycaemic excursions and the incidence of postprandial hypoglycaemia compared to currently available insulin analogues. LY900014 is an ultra-rapid-acting insulin lispro formulation that has shown an increased early absorption compared to commercially available insulin lispro (Humalog®; Eli Lilly and Company). LY900014 aims to closely mimic the physiological prandial insulin secretion pattern, which may more effectively control postprandial glucose excursions. FIASP® is a new commercially available fast-acting insulin aspart formulation that displays increased early absorption compared to commercially available insulin aspart (FIASP® SmPC, 2017; NovoRapid® SmPC, 2017).

This study in patients with type 1 diabetes mellitus (T1DM) aims to compare the pharmacokinetic (PK) profiles of insulin lispro after administration of LY900014 and Humalog and their impact on postprandial blood glucose during a standardised mixed meal tolerance test (MMTT). Furthermore, this study plans to characterise the PK and pharmacodynamics (PD) of insulin aspart analogues NovoRapid and FIASP in patients with T1DM under identical study conditions. A non-diabetic healthy subject cohort is included in the study to characterise "normal" insulin secretory and glucose response during an identical standardised test meal. Healthy subjects will not receive any of the study drugs during the study.

Evaluation of these meal-time insulins in the context of a standardised test meal cannot be used to infer clinical efficacy with the same rigour as a glycated haemoglobin (HbA1c) assessment following long-term treatment. However, assessment of these analogue insulin formulations under identical conditions provides a unique opportunity to obtain comparative information on the PK and PD in a setting appropriate for evaluation of meal-time insulin.

3.2. Background

The insulin analogue insulin lispro (Humalog) has been shown to be absorbed more quickly than regular human insulin (Humalog package insert, 2017). In healthy volunteers given subcutaneous (SC) doses of insulin lispro ranging from 0.1 to 0.4 units (U)/kg, peak serum levels were seen 30 to 90 minutes after dosing (Humalog package insert, 2015). However, the general consensus is that rapid-acting insulin is still not rapid enough to match carbohydrate absorption profiles, which limits efficacy and dosing flexibility. An ultra-rapid-acting prandial insulin would shift the PK/PD of insulin analogues so that they have an even faster onset to better match carbohydrate absorption and also allow greater flexibility in the time of dosing relative to meals.

LY900014 represents a new formulation that contains insulin lispro, treprostinil, citrate, and other excipients. This formulation involves the novel use of a microdose of treprostinil (CCI [redacted]) as an excipient to enhance the absorption of insulin lispro by local vasodilatation rather than as an active pharmaceutical ingredient to elicit a systemic effect. Treprostinil is a prostacyclin analogue, administered either through inhalation (CCI [redacted], as an intravenous (IV) infusion or as a continuous SC administration for the treatment of symptomatic pulmonary

arterial hypertension (PAH) and has been approved in the US since 2002 and in Germany since 2006 (AMIS database [WWW]). Sodium citrate is also included in the formulation to further enhance the absorption of insulin lispro, at least in part by enhancing vascular permeability. Each of the other excipients (such as magnesium chloride) in the LY900014 formulation is listed in the US Food and Drug Administration (FDA)'s Generally Recognized as Safe Food Additives database and in the FDA's Inactive Ingredients in Approved Drugs database. Furthermore, the excipient concentration in LY900014 is within the limits identified for approved drug products in the FDA Inactive Ingredients in Approved Drugs database.

To date, LY900014 has been administered to approximately 89 patients with diabetes (60 T1DM and 29 type 2 diabetes mellitus [T2DM]) across three Phase 1b studies and approximately 118 healthy volunteers across five Phase 1 studies. A further 24 patients with T1DM have received LY900014 in the ongoing CCI [redacted] Across these studies, LY900014 has consistently demonstrated a faster time-action profile than Humalog; however, the total insulin lispro exposure was similar between LY900014 and Humalog. In patients with T1DM, LY900014 reduced the time to early half-maximal drug concentration (early 50% t_{max}), a measure of insulin absorption, by 36.5% ($p<0.0001$) compared to Humalog. LY900014 displayed proportional increases in insulin lispro exposure (area under the concentration versus time curve [AUC] and maximum observed drug concentration [C_{max}]) with dose. The faster early insulin lispro absorption was maintained across the dose range studied (7.5 to 30 U). In patients with T1DM or T2DM treated with multiple daily injections (MDIs) of insulin therapy, LY900014 significantly reduced postprandial glucose excursions compared to Humalog when both were dosed through SC injection at the start of a standardised test meal. Additionally, through the use of MDIs of LY900014 for up to 2 weeks in patients with either T1DM or T2DM, it was found that LY900014 was well tolerated. There were no serious adverse events (SAEs) related to study treatment or discontinuations from the studies because of a drug-related adverse event (AE). Small numbers of treatment-emergent adverse events were reported, and there were no notable increases in these events in relation to any of the LY900014 formulations compared to those in relation to Humalog. There were no reported cases of severe hypoglycaemia related to LY900014 in these studies. Additionally, there were no reported incidences of local or systemic allergic reactions.

FIASP and NovoRapid (insulin aspart injection) are different formulations of the rapid-acting insulin analogue, insulin aspart, which when administered either SC or IV lowers blood glucose in patients with T1DM and T2DM. Insulin aspart is homologous with regular human insulin with the exception of a single substitution of the amino acid proline by aspartic acid at position B28, and is produced using recombinant DNA technology utilising *Saccharomyces cerevisiae*. In addition to the excipients in NovoRapid, FIASP also has arginine and nicotinamide as excipients (FIASP® SmPC, 2017; NovoRapid® SmPC, 2017).

3.3. Benefit/Risk Assessment

This study will not offer any direct benefits to the patients or healthy subjects participating in the study. The data from previous studies in healthy subjects and patients with T1DM and T2DM

have shown that LY900014 was well tolerated and the adverse drug reactions are in keeping with those reported for Humalog.

Potential risks associated with LY900014, derived from the known risks of insulin lispro (Humalog), are hypoglycaemia, hypersensitivity reactions (localised allergy and/or systemic allergy), undesirable effects at the injection site (injection-site reactions and lipodystrophy), and peripheral oedema (Humalog package insert, 2015).

Notably, across all doses in the studies that have evaluated treprostinil (CCI as a local vasodilator with or without insulin lispro, there was no clinically significant increase in those AEs associated with systemic absorption of treprostinil, as described in the CCI CCI The exposures of treprostinil in LY900014 for participants in ongoing and future clinical studies are expected to be much lower than those observed in the dose ranges previously explored with SC bolus administration of treprostinil. Accordingly, treprostinil exposure in diabetic patients is generally below the detection limit CCI and is substantially lower (at least ~200-fold lower) than those observed in adults for the treatment of PAH. No known potential risks are associated with the microdoses of treprostinil in the LY900014 formulation.

In preclinical safety pharmacology and toxicity studies, or clinical pharmacology studies involving LY900014 or treprostinil alone, other than known risks associated with Humalog and CCI no additional risks were identified. Additionally, local and systemic toxicity profiles of Humalog and Remodulin do not suggest the potential for additive or synergistic toxicity.

Potential risks associated with NovoRapid or FIASP, are hypoglycaemia, hypokalaemia hypersensitivity reactions (localised allergy and/or systemic allergy), undesirable effects at the injection site (injection-site reactions and lipodystrophy), weight gain and peripheral oedema (FIASP® SmPC, 2017; NovoRapid® SmPC, 2017).

The study includes inpatient procedures during which participants will be continuously monitored. In the event patients experience hypoglycaemia during the MMTT, then the hypoglycaemia event will be treated as described in Section 9.4.8. If a patient experiences elevated blood glucose level following food intake for more than 1 hour, insulin glulisine will be administered IV (see Section 9.4.8 for treatment guidelines). Patients will maintain their basal insulin regimen of site-provided insulin glargine during the entire study including the MMTT days. Appropriate measures will be taken to minimise the risk of hyperglycaemia (including conduct of a dose-finding assessment to ensure that the individualised dose is appropriate for the standardised test meal).

The healthy subjects in the study will not have any study drug intervention and therefore no potential risks associated with any of the study drugs are applicable.

Healthy subjects will undergo inpatient procedures (a single MMTT), during which participants will be continuously monitored.

More information about the known and expected benefits, risks, SAEs and reasonably anticipated AEs of LY900014 is to be found in the Investigator's Brochure (IB).

More detailed information about the known and expected benefits and risks of FIASP and NovoRapid may be found in the following: FIASP Summary of Product Characteristics and NovoRapid Summary of Product Characteristics.

4. Objectives and Endpoints

Table ITSL.1 shows the objectives and endpoints of the study.

Table ITSL.1. Objectives and Endpoints

Objectives	Endpoints
Primary	
1. To evaluate the differences in PK between LY900014 and Humalog following a single dose in patients with T1DM	1. Early 50% t_{max} and AUC(0-30min)
Secondary	
1. To characterise the PK profiles of LY900014, Humalog, NovoRapid and FIASP following a single dose in patients with T1DM	1. Early 50% t_{max}
2. To evaluate the differences in PD between LY900014 and Humalog following a single dose in patients with T1DM	2. ΔAUC(0-1h), ΔAUC(0-5h)
3. To characterise the PD response to LY900014, Humalog, NovoRapid and FIASP, as assessed by the MMTT, in patients with T1DM	3. ΔAUC(0-1h), ΔAUC(0-5h)
Tertiary/Exploratory	
1. To characterise "normal" insulin secretory and glucose response to the MMTT in non-diabetic healthy subjects	1. Insulin and glucose concentrations
2. To evaluate the tolerability of SC doses of LY900014 in patients with T1DM	2. AEs and hypoglycaemic events

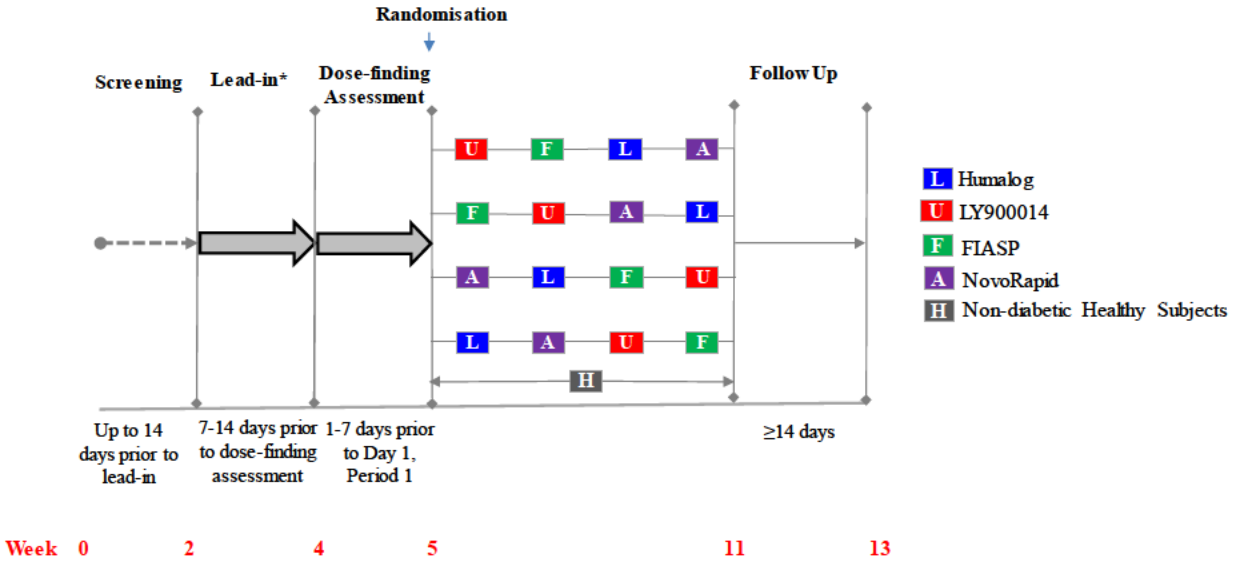
Abbreviations: AE = adverse event; AUC(0-30min) = area under the concentration versus time curve from time 0 to 30 minutes; ΔAUC = area under the baseline subtracted glucose concentration versus time curve; ΔAUC(0-1h) = ΔAUC from time 0 to 1 hour post-meal; ΔAUC(0-5h) = ΔAUC from time 0 to 5 hours post-meal; early 50% t_{max} = time to early half-maximal drug concentration; MMTT = mixed meal tolerance test; PD = pharmacodynamics; PK = pharmacokinetics; SC = subcutaneous; T1DM = type 1 diabetes mellitus.

5. Study Design

5.1. Overall Design

Study I8B-MC-ITSL (ITSL) is a clinical pharmacology, patient- and investigator-blind, randomised, 4-treatment, 4-period crossover study in patients with T1DM to compare insulin lispro PK and PD profiles of the postprandial blood glucose during a standardised MMTT following LY900014 in comparison to Humalog in patients with T1DM following single SC injections administered just before a standardised MMTT. This study will also characterise the PK and PD of insulin aspart analogues NovoRapid and FIASP in patients with T1DM under identical study conditions. A non-diabetic healthy subject cohort, which approximately matches to the cohort of patients with T1DM with regard to the mean body mass index (BMI) and age, is included with the intention of characterising the insulin secretory response to a standardised MMTT within normal physiology. These subjects will undergo a single in-house MMTT and will not receive any of the study drugs. Study ITSL may be conducted at 1 or more clinical research units (CRUs).

Figure ITSL.1 illustrates the study design.



Abbreviation: MMTT = mixed meal tolerance test.
*Lead-in period 7 to 14 days prior to dose-finding assessment to enable transition to insulin glargine; patients continue use of insulin glargine and usual prandial insulin analogues throughout the study during lead-in and between MMTT assessments.
Dose-finding MMTT with Humalog is performed between 1 and 7 days prior to MMTT in Period 1.
4-period crossover: the 4 MMTT assessments may occur on successive days, but all should be completed within 6 weeks.
Intravenous glucose insulin infusion is provided from 7 hours to 30 minutes prior to MMTT to achieve a stable baseline glucose target in the patients with type 1 diabetes mellitus.
Single MMTT is performed in non-diabetic healthy subject cohort; no lead-in period is required.

Figure ITSL.1. Illustration of study design for Protocol I8B-MC-ITSL.

5.1.1. Patients with T1DM

Each patient will be randomised on Day -1 of Period 1 to one of 4 treatment sequences (see Figure ITSL.1) comprising single SC doses of LY900014, Humalog, NovoRapid, and FIASP, administered immediately before the start of a standardised MMTT (time=0 min) (see Section 6.3.1).

Prior to Period 1, a dose-finding test will be performed to confirm that the patient's insulin to carbohydrate ratio is appropriate for the standardised MMTT for the study. The current study will use individualised dosing based on the dose-finding assessment, and this unit dose will be the same for each insulin (Humalog, LY900014, NovoRapid and FIASP) administration during the 4 MMTTs. The 4 MMTT assessments can occur on consecutive days but not more than 14 days apart, and all the 4 MMTT assessments should be completed within 6 weeks.

Patients will be required to visit the CRU on at least 7 occasions (no more than 11 occasions if dose finding and/or 1 MMTT are rescheduled) as noted in the Schedule of Activities (see Section 2):

- Screening visit
- Lead-in (insulin transition) period (see Section 5.1.1.1)
- 4 inpatient CRU study visits (including dose-finding assessment [see Section 5.1.1.3])
- A follow-up visit at least 14 days after the last dose or early discontinuation

Eligible patients who have fulfilled the entry criteria and completed all screening procedures will return to the CRU 7 to 14 days prior to the dose-finding assessment visit to begin a lead-in (insulin transition) period. When patients visit the CRU to begin the lead-in period, patients will receive and review instructions on general diabetes education including measurement of self-monitored plasma glucose (SMPG), and on the insulin transition (see Section 9.4.7). During the insulin transition period, patients will transition from their current basal insulin therapy to the CRU-provided insulin glargine (see Section 5.1.1.1). Following insulin transition, patients will return to the CRU for dose-finding assessment (see Section 5.1.1.2) from 1 to 7 days prior to Day 1 of Period 1. The dose-finding assessment will involve the use of Humalog administered immediately (not more1 minute) before a standardised MMTT to inform the dose of prandial insulin (LY900014, Humalog, FIASP, or NovoRapid) to be used during the study MMTT assessments. Upon completion of the dose-finding assessment, patients will be randomised to 1 of 4 treatment sequences. Prior to the MMTT in each study period, patients will undergo a run-in period (see Section 5.1.1.3) to achieve a predetermined glucose target of 135±15 mg/dL (7.5±0.8 mmol/L). Once the glucose target is achieved, patients will proceed with the MMTT in which a single individualised SC dose of LY900014, Humalog, FIASP, or NovoRapid will be administered immediately before the standardised test meal (see Section 5.1.1.3).

Each dose of study drug (LY900014, Humalog, FIASP, and NovoRapid) will be separated by a minimum of 21 hours during all the treatment periods. The maximum duration allowed for all 4 treatment periods is approximately 6 weeks. Patients will continue using CRU-provided insulin glargine and their usual rapid- or fast-acting insulin analogues (prandial insulin) during the entire study including the lead-in period and between MMTTs (see Section 7.7 for use of concomitant medications and basal insulin). During the washout periods, patients will be instructed to perform regular monitoring of blood glucose (see Section 9.4.7).

Following completion of all study procedures in each period, patients may be offered a meal and discharged from the CRU at the discretion of the investigator.

5.1.1.1. Lead-in and Insulin Transition

After completing all screening procedures according to the Schedule of Activities (Section 2), patients will return to the CRU for the lead-in visit and receive instructions on the insulin transition. The lead-in CRU visit is to take place 7 to 14 days prior to the dose-finding assessment. At the lead-in CRU visit, patients will transition from their current regimen of basal insulin to the CRU-provided insulin glargine according to the following guidance:

- 7 to 14 days prior to their admission to the CRU for the dose-finding assessment, patients will switch from neutral protamine Hagedorn (NPH) insulin, insulin detemir, insulin degludec or insulin glargine to the CRU-provided once-daily evening dose of insulin glargine. The starting dose will be at the investigator's discretion using patients' current insulin regimen as the reference.
- Adjustments may be made in consultation with the investigator following feedback obtained from the patient during at least twice-weekly telephone follow-up as part of the insulin-transition period.
- Once a stable dose has been established, patients will remain on it till completion of the study.

Patients can continue current therapy with any prandial insulin throughout the study.

Additionally, at the lead-in CRU visit, patients will be provided with general diabetes training including, but not limited to proper insulin-injection technique, correct self-monitoring of plasma glucose (PG) using a standardised glucometer, interpretation of results and symptoms, and treatment of hypoglycaemia. In addition to the study insulin, the CRU will also provide the glucometer, test strips, control solution and lancets. A diary will also be provided as described in Section 9.4.7.

5.1.1.2. Dose-Finding Assessment

After transitioning to a stable dose of insulin glargine, patients will return to the CRU anytime between 1 and 7 days prior to MMTT in Period 1 for the dose-finding assessment. The dose-finding assessment may begin between 0700 and 1100 hours. Patients will have fasted for at least 10 hours prior to the standardised test meal (see Section 5.1.1.3). The dose-finding test will be performed to confirm that the patient's insulin to carbohydrate ratio is appropriate for the standardised MMTT for the study. The objective of the dose-finding assessment is to determine an appropriate, individualised prandial insulin dose to be administered with the standardised test meal during the study MMTT (see Section 5.1.1.3). The prandial insulin selected from the dose-finding assessment will be same for all insulins (Humalog, LY900014, NovoRapid and FIASP) assessed within the study.

The patient's blood glucose level will be measured to select a Humalog dose for the dose-finding assessment as follows:

- if the patient's blood glucose level is within the range of 70 to 180 mg/dL (3.9 to 10.0 mM), the patient will be given Humalog at the dose the patient would have taken at home
- the dose-finding meal test should not be carried out if the patient's blood glucose level is not within the range of 70 to 180 mg/dL (3.9 to 10.0 mM) prior to start of the dose-finding meal test. The dose-finding meal test may be rescheduled as judged by the investigator (see Section 6.2.3).

Immediately (within 1 minute) after the Humalog dose has been selected and administered, patients will receive a standardised test meal as described in Section 6.3.1. Patients are expected to complete each standardised test meal within 15 minutes of starting it.

The patient's blood glucose level will be measured every 20 minutes for 5 hours post-meal to select a Humalog dose for the study MMTT as follows:

- if the patient's blood glucose level is within the range of 70 to 240 mg/dL (3.9 to 13.3 mM) for 5 hours post-meal, the selected Humalog dose will be used for the prandial insulin dose (LY900014, Humalog, FIASP and NovoRapid) to be administered along with the MMTTs in all the periods but can still be adjusted if necessary based on the clinical judgement of the investigator.
- if the patient's blood glucose level is not within the range of 70 to 240 mg/dL (3.9 to 13.3 mM), the dose of Humalog will be adjusted based on the investigator's judgement or if the investigator is unable to make a decision regarding the Humalog dose after completing the initial test, the dose-finding assessment may be repeated once within 7 days of the initial assessment.
- if, during the dose-finding assessment, the patients' blood glucose level drops below pre-test baseline, the investigator may make further adjustments of the insulin dose. Refer to Section 9.4.8 for treatment of hyperglycaemia and hypoglycaemia.

Following completion of procedures associated with the dose-finding assessment, patients may be offered a meal. Dose-finding assessment should occur between 1 and 7 days prior to MMTT in Period 1. If the dose-finding assessment occurs from Day -2, the patient may continue to remain inpatient in the CRU until the completion of study MMTT on Day 1 of Period 1.

5.1.1.3. Inpatient Procedures

Run-In

On the day before each MMTT day (that is, Day -1 in each of the 4 periods), patients will arrive at the CRU in the early evening (approximately 1700 hours) except for patients who undergo the dose-finding assessment from Day -2 of Period 1. Patients will be advised to bring their prandial insulin when they arrive at the CRU, which will be administered before the start of a standardised dinner.

Before the start of the run-in period, cannulation of 2 veins will be performed. A variable IV infusion of glucose (20% dextrose solution) and/or insulin glulisine will be initiated (~2400 hours) in order to obtain a blood glucose target level of 135±15 mg/dL (7.5±0.8 mmol/L). Any insulin infusion should be tapered off and stopped at least 30 minutes prior to start of the standardised test meal. For the last 30 minutes prior to study drug administration, the target blood glucose concentration should be within the range without any glucose infusion. If this target blood glucose range is not attained before 1100 hours, the meal test will be halted and may be performed on a separate visit; each meal test can only be repeated once (see Section 6.2.3 for rescheduling).

Study MMTT

Each patient's individual, short-acting insulin may be administered on Day -1 before the start of a standardised dinner (at approximately 1900 hours). If a correction dose of short-acting insulin needs to be administered after the meal, no more than 6 U of short-acting insulin should be administered between 7 and 12 hours before the scheduled intake of standardised test meal.

The meal test may start in the early morning at approximately 0700 hours with allowance up to 1100 hours, if required, to ensure the patient's blood glucose level is stable and on target, with the pre-meal activities as specified in the Schedule of Activities (Section 2). LY900014, Humalog, FIASP, or NovoRapid SC bolus injection will be administered immediately before the start of MMTT during each of the 4 treatment periods according to the randomisation schedule. The individualised dose for each patient will be the same for all the 4 treatment periods. For each MMTT, the patient should stay in a semi-supine position for 2 hours post-meal. The patient will not be allowed to consume water for 2 hours after the MMTT assessment begins; however, water may be consumed freely 2 hours post-meal.

A standardised MMTT will be administered on Day 1 during each of the 4 treatment periods. Patients are expected to complete each standardised test meal within 15 minutes of starting it. Patients will be without further food intake from the start of the meal to completion of blood collection (approximately 420 minutes) unless required to treat hypoglycaemia (see Section 9.4.8).

Following completion of the assessment period and sample collection (approximately 420 minutes), patients will be offered a meal. Patients will continue with the CRU-provided insulin glargine along with their usual bolus prandial insulin. Patients will be observed following administration of the meal and discharged at the investigator's discretion.

After completion of the last treatment period (Period 4), the patients will resume their pre-study basal insulin.

5.1.2. Healthy Subjects

The cohort of healthy subjects will be approximately matched to the cohort of patients with T1DM as far as possible with regard to the mean BMI and age. The healthy subjects will be assessed for the PK and PD effects of their endogenous insulin. They will not be administered any study drugs, but will undergo MMTT as described below.

The subjects will be required to visit the CRU on at least 3 occasions as noted in the Schedule of Activities (see Section 2):

- Informed consent
- Screening visit (including an oral glucose tolerance test [Appendix 6]) up to 28 days prior to MMTT
- An inpatient CRU study visit

Eligible healthy subjects who have satisfied the entry criteria and completed all screening procedures will return to the CRU on Day -1 of the inpatient visit. Subjects will be offered a standardised dinner (at approximately 1900 hours).

On Day 1, the MMTT may start at approximately 0700 hours, with the pre-meal activities as specified in the Schedule of Activities (Section 2). A standardised MMTT will be performed on Day 1 and the subjects are expected to complete the standardised test meal within 15 minutes of starting it. The subjects should stay in a semi-supine position for 2 hours post-meal. The subjects will not be allowed to consume water for 2 hours after the MMTT assessment begins; however, water may be consumed freely 2 hours post-meal.

Subjects will be without further food intake from the start of the meal to completion of blood collection (approximately 420 minutes) unless required to treat hypoglycaemia (see Section 9.4.8).

Following completion of the assessment period and sample collection (approximately 420 minutes), subjects will be offered a meal. Subjects will be observed following administration of the meal and discharged at the investigator's discretion.

Study governance considerations are described in detail in Appendix 3.

5.2. Number of Participants

Up to 74 patients with T1DM may be enrolled so that approximately 64 patients complete the study. Replacement patients will adopt all 4 assigned crossover treatments of the original patient's randomisation schedule.

For the healthy subject cohort, sufficient subjects will be enrolled to have 12 healthy subjects complete the study.

For purposes of this study, a patient/subject completes the study when all scheduled procedures shown in the Schedule of Activities (see Section 2) have been finished.

5.3. End of Study Definition

End of the study is the date of the last visit or last scheduled procedure shown in the Schedule of Activities (Section 2) for the last patient/subject.

5.4. Scientific Rationale for Study Design

This study has a 4-period crossover design to reduce the variability of insulin PK and PD as each patient will serve as his/her own control. Also, the total number of patients needed with a crossover design is less than the number needed with a parallel group design. A maximum duration of approximately 6 weeks is allowed for patients to complete all 4 assigned periods to minimise the risk of insulin resistance/changes in mean glycaemic control during the study.

This study is being conducted in patients with T1DM, as this is the target population for the study drugs. Additionally in patients with T1DM there is no interference of endogenous

insulin. A healthy subject cohort is being included to characterise the normal physiological insulin secretory response to a single standardised MMTT.

Randomisation and blinding are used to avoid bias introduced through an association between allocation order of study drugs and patient characteristics. The Lilly clinical pharmacologist (CP)/Lilly study team will be unblinded.

In each study period, patients will undergo a run-in period before the MMTT using a variable insulin and glucose IV infusion. This run-in period will allow for improved comparability of the postprandial glucose response to a mixed meal after treatment with LY900014, Humalog, FIASP, and NovoRapid when administered immediately before the start of the meal. This run-in aims to achieve similar preprandial glucose levels for all patients before the start of the standardised test meal and thereby reduces the variability of the postprandial glucose response. Insulin glulisine has been chosen for IV optimisation of blood glucose during the run-in because it does not cross-react with either the insulin lispro- or insulin aspart-specific assay used for the PK analysis.

Under this design, if 2 periods occur on consecutive days, the interval between the last bolus on the first day and the first bolus on the second day is much longer compared to the length of time that the treatment (LY900014, Humalog, FIASP or NovoRapid) lasts in the bloodstream; therefore, no carryover effect is expected. This enables PK and PD data from the breakfast meal tests of each period to be analysed independently and separately.

At least 21 hours between the doses of study drug allows for a complete washout of study drug administered with the MMTT and glucose response and prevents carryover effects.

5.5. Justification for Dose

The bolus dose of insulin (LY900014, Humalog, FIASP, or NovoRapid) will be individualised to cover the carbohydrate content in this standardised test meal. This dose of insulin is reflective of clinically relevant, individualised insulin dosing, similar to the manner in which patients would determine the insulin dose for the carbohydrate content in a meal. For each patient, the individualised prandial insulin dose for each standardised test meal must be kept identical throughout the crossover periods.

6. Study Population

Eligibility of patients or subjects for the study will be based on the results of screening medical history, physical examination, vital signs, clinical laboratory tests and electrocardiogram (ECG).

The nature of any conditions present at the time of the physical examination and any pre-existing conditions will be documented.

Screening for patients may occur up to 14 days prior to the lead-in period. Patients who are not enrolled within 35 days of screening may be subjected to an additional medical assessment and/or clinical measurements to confirm their eligibility.

Screening for healthy subjects may occur up to 28 days prior to Day 1. Subjects who are not enrolled within 28 days of screening may be subjected to an additional medical assessment and/or clinical measurements to confirm their eligibility.

Prospective approval of protocol deviations to recruitment and enrolment criteria, also known as protocol waivers or exemptions, is not permitted.

6.1. Inclusion Criteria

Patients or subjects are eligible for inclusion in the study only if they meet all of the following criteria at screening and/or enrolment:

[1] For patients with T1DM: are male or female patients with T1DM for at least 1 year. A diagnosis of T1DM is based on medical history with a fasting C-peptide level <0.30 nmol/L.

for the non-diabetic healthy subjects: are overtly healthy males or females with normal glucose tolerance (see Section 6.1.2), as determined through medical history and physical examination.

a. Male patients:

i. No male contraception required except in compliance with specific local government requirements.

b. Female patients:

i. Women of child-bearing potential who are abstinent (if this is complete abstinence, as their preferred and usual lifestyle) or in a same sex relationship (as part of their preferred and usual lifestyle) must agree to either remain abstinent or stay in a same sex relationship without sexual relationships with males.

ii. Otherwise, women of child-bearing potential participating in the study must agree to use 1 highly effective method of contraception until discharge from final treatment period.

1. Women of child-bearing potential must test negative for pregnancy prior to initiation of treatment as indicated by a negative serum pregnancy test at the screening visit followed by a negative urine pregnancy test within 24 hours prior to exposure in Period 1.

2. A highly effective method of contraception includes a combined (oestrogen and progestogen containing) or progestogen-only hormonal contraception administered orally, intravaginally or transdermally and is associated with inhibition of ovulation. Alternatively, patients may use either an intrauterine device, intrauterine hormone-releasing system, bilateral tubal occlusion, or the partner should have been vasectomised.

iii. Women not of child-bearing potential may participate and include those who are

1. infertile due to surgical sterilisation (hysterectomy, bilateral oophorectomy or bilateral salpingectomy), congenital anomaly such as Mullerian agenesis; or

2. post-menopausal – defined as a woman being amenorrhoeic for more than 1 year without an alternative medical cause and a serum follicle-stimulating hormone (FSH) level compatible with post-menopausal status. An FSH level in the post-menopausal range may be used to confirm a post-menopausal state in women not using hormonal contraception or hormonal replacement therapy.

[2] are aged 18 to 70 years, both inclusive.

[3] have a BMI of 18.5 to 30.0 kg/m^2, both inclusive.

[4] have clinical laboratory test results within normal reference range for the population or investigative site, or results with acceptable deviations that are judged to be not clinically significant by the investigator.

[5] have venous access sufficient to allow for glucose infusion and blood sampling as per the protocol.

[6] are reliable and willing to make themselves available for the duration of the study and are willing to follow study procedures including the ability to consume the standardised test meal.

[7] are able and willing to give signed informed consent.

6.1.1. Additional Inclusion Criteria for T1DM

[8] have an HbA1c ≤9.0% at screening.

[9] have had no episodes of severe hypoglycaemia in the past 6 months (see Section 9.4.6.2).

[10] are on stable prandial insulin and basal insulin (NPH insulin, insulin glargine or insulin detemir or insulin degludec) for at least 3 months before screening with a total insulin dose demand of ≤1.5 U/kg/day.

6.1.2. Additional Inclusion Criteria for Healthy Subjects

[11] have a fasting PG of ≤100 mg/dL, and post-challenge PG of ≤140 mg/dL following oral glucose tolerance test (Appendix 6) and HbA1c <5.7% at screening.

6.2. Exclusion Criteria

Patients or subjects will be excluded from study enrolment if they meet any of the following criteria at screening and/or enrolment:

[12] are CRU personnel directly affiliated with this study and their immediate families. Immediate family is defined as a spouse, biological or legal guardian, child or sibling.

[13] are Lilly employees or are employees of the CRU.

[14] are currently enrolled in a clinical study involving an investigational product or any other type of medical research judged not to be scientifically or medically compatible with this study.

[15] have participated, within the past 30 days or 5 half-lives (whichever is longer) in a clinical study involving an investigational product.

[16] have previously completed or withdrawn from this study.

[17] have an abnormality in the 12-lead ECG that, in the opinion of the investigator, increases the risks associated with participating in the study.

[18] have a supine blood pressure at screening outside the range of 90 to 160 mmHg for systolic or 50 to 100 mmHg for diastolic (1 repeat is allowed) as determined by the investigator, or results with unacceptable deviations that are judged by the investigator to be clinically significant for the population, or have a heart rate outside the range of 50 to 90 beats/minute.

[19] have known or ongoing psychiatric disorders.

[20] regularly use known drugs of abuse and/or show positive findings on urinary drug screening.

[21] show evidence of an acute infection with fever or infectious disease at the time of study entry.

[22] show evidence of human immunodeficiency virus (HIV) infection and/or positive human HIV antibodies.

[23] show evidence of hepatitis C and/or positive hepatitis C antibody (the presence of hepatitis C antibodies in the setting of normal liver function tests and a negative hepatitis C polymerase chain reaction are not an exclusion).

[24] show evidence of hepatitis B and/or positive hepatitis B surface antigen (the presence of antibodies to the hepatitis B surface antigen is not an exclusion).

[25] are women who are pregnant or lactating.

[26] have a history of renal impairment (exclusion only if estimated glomerular filtration rate [GFR] <60 mL/minute/1.73 m^2 [GFR is estimated according to the Chronic Kidney Disease Epidemiology Collaboration creatinine equation], or have a serum creatinine level ≥126 µmol/L (male) or ≥111 µmol/L (female).

[27] have a history of deep vein thrombosis of the leg or repeated episodes of deep leg vein thrombosis in first-degree relatives (parents, siblings or children) as judged by the investigator.

[28] are currently smoking more than 5 cigarettes (or nicotine equivalent) per day.

[29] have donated blood of more than 450 mL or more in the past 3 months or provided any blood donation within the last month before screening.

[30] have a significant history of alcoholism or drug abuse as judged by the investigator.

[31] currently consumes more than 24 g of alcohol per day for men, or more than 12 g of alcohol per day for women (1 unit of alcohol is defined as 10 mL [8 g] of pure alcohol).

[32] are unwilling to comply with the dietary requirements/restrictions during the study: (i) comply with the fasting requirements of the study, (ii) consume only the meals/snacks provided during the inpatient visits.

[33] are receiving chronic (lasting longer than 14 consecutive days) systemic or inhaled glucocorticoid therapy (excluding topical, intra-articular and intraocular preparations), or have received such therapy within 4 weeks before screening.

[34] in the opinion of the investigator or sponsor, are unsuitable for inclusion in the study.

6.2.1. Additional Exclusion Criteria for T1DM

[35] have significant lipohypertrophy in the target abdominal injection area as judged by the investigator.

[36] patients with T1DM having proliferative retinopathy or maculopathy and/or severe neuropathy; in particular, autonomic neuropathy as judged by the investigator based on a recent (<1.5 years) ophthalmologic examination.

[37] have a history or current cardiovascular, respiratory, hepatic, renal, gastrointestinal, haematological, or neurological disorders capable of significantly altering the absorption, metabolism, or elimination of drugs; of constituting a risk when taking the investigational product; or of interfering with the interpretation of data.

[38] have, except for the current regimen of insulin therapy and concomitant medication(s) (for example, antihypertensive medication, lipid-lowering agent, thyroid hormone replacement medication, hormonal contraception, hormonal replacement therapy), regular use of or intended use of any over-the-counter or prescription medications or nutritional supplements that promote weight loss within 14 days before dosing.

[39] have type 2 diabetes and use other injectable or oral medications to treat hyperglycaemia (such as GLP-1 receptor agonists, metformin, or SGLT2 inhibitors)

[40] have known allergies to treprostinil (CCI [redacted], insulin lispro, insulin glulisine, insulin glargine, FIASP, insulin aspart-related compounds or any components of the formulation, or a history of significant atopy.

[41] any significant changes in insulin regimen and/or unstable blood glucose control within the past 3 months prior to screening as assessed by the investigator.

6.2.2. Additional Exclusion Criteria for Healthy Subjects

[42] have a history or current cardiovascular, respiratory, hepatic, renal, gastrointestinal, endocrine, haematological, or neurological disorders capable of significantly altering the absorption, metabolism, or elimination of drugs; of constituting a risk when taking the investigational product; or of interfering with the interpretation of data.

[43] are healthy nondiabetic volunteers that currently use any medications that impact insulin secretion or elevate glucose (such as beta blockers, thiazide diuretics, and niacin)

6.2.3. Additional Exclusion Criteria for Inpatient Dosing Days

Patients or subjects who fulfil 1 or more inpatient dosing-day exclusion criteria will be excluded from study drug administration for that MMTT day. A single inpatient treatment period can be rescheduled 1 to 14 days later. Each treatment period can only be rescheduled once.

The following exclusion criteria apply to the day before each MMTT day:

[44] any medical condition or AE that could interfere with glucose metabolism, as judged by the investigator.

[45] during the run-in period and up to 2 hours prior to the start of the MMTT, episodes of non-severe hypoglycaemia (symptoms or BG ≤70 mg/dL [3.9 mmol/L]) can be treated with 15 to 20 g of carbohydrate. If a hypoglycaemia episode requires more than approximately 20 g of carbohydrate within 8 hours of the start of the MMTT or the patient experiences a severe hypoglycaemia episode up to 1 week prior to MMTT (as defined in Section 9.4.6.2 and 9.4.6.3), the MMTT must be rescheduled.

[46] bolus injection of more than 6 U of a prandial insulin between 7 and 12 hours before standardised test meal.

[47] any use of caffeine-containing food/beverages (for example, cola, chocolate drinks, tea, coffee, energy drinks containing methylxanthine [caffeine, theophylline, or theobromine]) for at least 12 hours before each standardised test meal and throughout the duration of each CRU visit.

6.3. Lifestyle and/or Dietary Requirements

Throughout the study, patients and subjects may undergo medical assessments and review of compliance with requirements before continuing in the study.

6.3.1. Meals and Dietary Restrictions

Patients and subjects will be provided standardised liquid meals for the MMTTs, as outlined in the Schedule of Activities (Section 2). A standardised evening meal on the day prior to the MMTT and a meal following completion of the MMTT will also be provided. Patients and subjects will be fasted (except for water) for at least 10 hours before each standardised test meal and consume each meal within approximately 15 minutes. The standardised test meal will use CCI ®. The standardised test meal will consist of 2 cans of 8 fl/oz CC CI (approximately 100 g of carbohydrates total in each standardised test meal). Patients and subjects will not be allowed to consume water for 2 hours after consuming the standardised test meal apart from the fluid provided with the meal; however, water may be consumed freely afterwards.

While resident in the CRU, patients and subjects may not consume any food or caloric drinks other than that provided by the CRU personnel. When not resident in the CRU, patients and subjects may resume their regular diet.

Following completion of study procedures, patients and subjects will be offered a meal. The CRU personnel will record dose and time of prandial insulin administered to patients with T1DM with end-of-study meal.

6.3.2. Caffeine, Alcohol and Tobacco

Patients and subjects should refrain from caffeine-containing food/beverages (for example, cola, chocolate drinks, tea, coffee, energy drinks containing methylxanthine [caffeine, theophylline or theobromine]) for at least 12 hours before each standardised test meal and throughout the duration of each CRU visit.

No alcohol will be allowed for at least 24 hours before each CRU admission (Day –1) and throughout the duration of each CRU visit. Between CRU visits, daily alcohol should not exceed 3 units for males and 1.5 units for females (see Section 6.2, [31] for unit definition).

No tobacco products including cigarette, cigars, pipes and nicotine replacement will be permitted during the inpatient visits.

6.3.3. Activity

Patients and subjects will be encouraged to maintain their regular exercise and insulin regimen adaptation (for patients with T1DM only) during the outpatient period; however, they should not undertake vigorous or prolonged exercise at least 24 hours before each dosing day at the CRU. While in-patient, movement will be restricted to retain the integrity of connections to infusion(s) and the study procedures.

6.4. Screen Failures

Individuals who do not meet the criteria for participation in this study (screen failure) may not be re-screened.